

6th International Conference on Industry 4.0 and Smart Manufacturing

Literature survey on combining machine learning and metaheuristics for decision-making

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Abstract

The scope and complexity of decision-making in modern smart manufacturing, coupled with the precision and performance requirements, lead to widespread reliance on various computational solutions based on machine learning or metaheuristics. In this work, we perform a systematic literature review of approaches that propose the combined use of both of the aforementioned algorithm types to assist decision-makers in various fields. In total, 20 relevant publications were identified. Out of these, 17 publications represent supervised and unsupervised machine learning algorithms, both classical and algorithms based on neural networks, that are employed in combination with metaheuristics for decision-making. However, ten out of the proposed approaches apply metaheuristics merely for the parameterization of machine learning algorithms or feature selection. Only three articles have been discovered that rely on the principles of reinforcement learning. Out of all of the relevant publications reviewed, only four focus on tackling the challenges of decision-making in manufacturing, and a single proposes a solution in logistics. The rest represent approaches in various other domains.

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Peer-review under responsibility of the scientific committee of the 6th International Conference on Industry 4.0 and Smart Manufacturing

Keywords: Decision-support; Smart Manufacturing; Metaheuristics; Machine Learning;

1. Introduction

The concept of smart manufacturing [1] relies on a variety of complex monitoring, planning, automation, cloud, and data-driven solutions. In fact, optimized decision-making is one of the core characteristics of smart factories [2]. The introduction of complex sensor data collection and processing, automation, and management solutions offers the possibility of making faster and more accurate decisions that reflect the changes in supply chains or manufacturing processes. This decision-making process is supported by automation and optimization performed at various stages of the production process to achieve the desired result within a reasonable time scope.

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Nowadays, machine learning is commonly applied in a variety of decision-making tasks directly related to smart manufacturing [3]. At the same time, numerous metaheuristic algorithms found their application in manufacturing optimization and planning tasks [4]. There are also attempts to combine metaheuristics and machine learning in a smart manufacturing scenario. For example, a recent publication [5] outlines the use of metaheuristics can be used as a training input to a deep reinforcement learning algorithm for production scheduling while using a hybrid shop floor simulation for evaluation. It is also shown in the literature[6], that it is possible to combine metaheuristics and machine learning algorithms into a hybrid solution for scheduling in the manufacturing context.

It is clear that combining various types of algorithms may be beneficial for decision-making in smart manufacturing. In this work, we strive to create a brief overview of the published literature where various algorithms are combined together as part of a decision-making process. Specifically, we focus our attention on the combined use of metaheuristic algorithms and machine learning algorithms. The goal of this work is to analyze the published algorithm combinations and their application in decision-making. Therefore, the analysis is not limited to only smart manufacturing and logistics but also includes general use cases. Investigating the broader application scope allows us to analyze the similarities and differences between approaches specific to manufacturing and those from other fields of study or industries.

In this work, we conduct a systematic literature review of publications focusing on decision support, which relies on a combination of metaheuristics and machine learning. To the best of our knowledge, there are no recent published reviews that focus on decision-making with the help of combining metaheuristic and machine learning algorithms. We attempt to gain an understanding of which algorithms are used in such a combination for decision support and in which manner they are combined within the published literature.

In our work, we differentiate three types of machine learning. First, there is traditional machine learning. Algorithms in this category rely on principles from probability theory and statistics [7] to process data and make predictions or perform classification based on it. These are supervised, unsupervised, or semi-supervised algorithms that do not rely on artificial neural networks or reinforcement learning principles.

A major part of the modern research in the area of machine learning concentrates on artificial neural networks. Therefore, we can not ignore this area in this work. Generally, two major types of neural networks in machine learning are differentiated [8]: artificial neural networks (ANN) and deep neural networks (DNN). The main difference between the two is the number of layers of neurons. Specifically, ANN has only a single layer between input and output. DNN, however, typically consists of multiple layers of neurons situated between input and output. These layers are often referred to as hidden layers. Important to note ANN and DNN-based algorithms in this context include supervised and unsupervised learning, similar to the aforementioned traditional machine learning category.

The last type of machine learning we consider is reinforcement learning (RL) [9]. This includes deep reinforcement learning (DRL) [10], a subset of RL that relies on deep neural networks. This subset of machine learning differs significantly from the one discussed above as it does generally not require data for training but instead is provided with a so-called environment where an instance of the algorithm, also called an agent, can perform some actions within a predefined set of possible actions. Therefore, RL is commonly separated into its own category of machine learning approaches, which is different from already mentioned supervised and unsupervised machine learning.

2. Research Methodology

In this section, we outline the systematic process of searching and selecting the published literature relevant to this research. Specifically, the scope of this work concentrates on peer-reviewed publications outlining a combination of machine learning and metaheuristics in the decision-making process.

We relied primarily on Scopus¹ for literature search. However, in order to broaden the pool of discovered literature, we also performed a search directly in the following eight databases of the published scientific research: Springer²,

¹ <https://www.scopus.com/>

² <https://link.springer.com/>

IEEE³, ACM⁴, Sage Journals⁵, NIPS⁶, Emerald insight⁷, Taylor & Francis⁸, Science Direct⁹. Furthermore, a forward and backward search is also performed for selected papers, further expanding the search field. The search is constrained to the literature published between 2008 and 2024 in the English language. We concentrate on conference publications and journal articles. Books, as well as Master's and PhD theses, are not included in the search. Only the works with fully accessible text are considered for further analysis.

Specific keywords used in the search process are listed in Table 1. A variety of spelling of the terms is included to cover the different possible ways of naming the same concept in the literature. The keyword groups are combined using AND logical conjunctions according to the aforementioned goals of the literature review.

Table 1. Search keywords.

Group	Search Keywords
Decision framework	("decision making" OR "decision support" OR "decision science" OR "decision framework")
Machine Learning	("machine learning" OR "artificial intelligence" OR "ai" OR "deep learning" OR "rl" OR "reinforcement learning" OR "drl" OR "deep reinforcement learning" OR "deep rl")
Metaheuristics	("metaheuristic" OR "meta-heuristic" OR "meta heuristic")

Furthermore, the literature is selected from the search results in multiple steps. In the first step, the selection is based on the title. Publications with an irrelevant title are discarded. After that, we further prune the selected literature based on the relevance of the abstract for the scope of our research. Finally, the remaining publications are analyzed in detail. Initially, 474 papers were discovered based on the search keyword combinations, out of which only 124 were deemed relevant based on the title. After reading the abstract, only 38 papers were selected for full reading, out of which just 20 were deemed relevant for this analysis and presented in this work.

3. Literature review

In this section, we briefly present the relevant literature that was discovered. The section begins with subsection 3.1 where we present the literature where authors propose a combination of a metaheuristic and traditional machine learning. Then, the section continues with subsection 3.2, where publications that bring the neural-network-based approaches and metaheuristics together in a decision-making process are discussed. It is important to note that, while some publications fall in both aforementioned categories, the summaries for such are presented only in subsection 3.1 to avoid repetition. Following this, in subsection 3.3, the approaches that are based on the principles of reinforcement learning combined with metaheuristics are discussed. The section finally concludes in subsection 3.4 with a high-level overview of all of the discovered literature and a short discussion.

3.1. Traditional machine learning and metaheuristics

Half of all discovered publications constitute publications presenting a combination of traditional machine learning and metaheuristics. It makes it the most numerous algorithm category out of the three defined in our work.

An example of using metaheuristics for selecting optimal parameters for machine learning is proposed [11] in a solution-oriented on researchers and decision-makers in the areas of disaster management and prediction using machine learning algorithms. Specifically, the authors rely on Least-squares support-vector machines (LS-SVM) machine

³ <https://ieeexplore.ieee.org/>

⁴ <https://dl.acm.org/>

⁵ <https://journals.sagepub.com/>

⁶ <https://papers.nips.cc/>

⁷ <https://www.emerald.com/insight/>

⁸ <https://taylorandfrancis.com/journals/>

⁹ <https://www.sciencedirect.com/>

learning algorithm used for regression. The authors note the critical role of selecting the right values for the parameters of the algorithm. To tackle this challenge, the authors apply a metaheuristic algorithm to search for the optimal parameters. Specifically, authors employ a modified version of a firefly algorithm (FA) [12] within the scope of their work.

A further publication combining traditional ML and metaheuristics, presented in the area of healthcare [13] and aimed at supporting the decision-making in early diagnosis of diabetes. The authors rely on the XGBoost [14] machine learning algorithm for classification, while a non-dominated sorting genetic algorithm (NSGA-II) [15] is used for selecting the features used to train the XGBoost model in order to maximize its performance.

In contrast to previous examples, the authors of a publication in the domain of logistics [16] propose a pipeline of data preparation, machine-learning-based regression analysis, and particle swarm optimization (PSO) [17] for the optimization of speed selection based on the wealth conditions for the transport ships. The authors present the solution relying on the historical data representing naval travel and weather conditions, which is used to train a regression model that presents the estimation of fuel consumption given weather conditions. Then a metaheuristic is used to propose a steering and seed solution based on that estimation.

The authors of a recent publication [18] rely on a modified Hitchcock bird-inspired algorithm (HBIA) [19] metaheuristic and K-Nearest-Neighbors (KNN) machine learning algorithm to provide decision support in the deployment of a wireless sensor network. The authors name their modification of HBIA algorithm I-HBIA. The authors rely on the chosen metaheuristic to select the optimal placement of the sensor network nodes. KNN trained on a real dataset is used to estimate the strength of the signal, which is required for the evaluation of the optimization solutions provided by HBIA.

A further example of applying metaheuristics for maximizing the performance of a machine learning algorithm is proposed [20] with a focus on selecting only the information from the training dataset that is beneficial for training the chosen machine learning model. The authors rely on heuristic and metaheuristic methods to select features in the dataset describing the employee turnover factors. The selected features are then used for training machine learning models to make a prediction on employee turnover. Specifically, authors rely on the gray wolf optimizer [21] and genetic algorithm metaheuristics. After that, the dataset consisting of the features selected by the metaheuristics is used to train the following popular traditional ML models: logistic regression (LR), decision tree (DT), random forest (RF), gradient boosting (GB), support vector machines (SVM), Adaptive Boosting (AdaBoost). However, the aforementioned algorithms are not all used together. Instead, the authors compare the performance of the metaheuristic and machine learning pairs individually.

A combination of multiple various algorithms is presented [22] for incident management. The authors rely on a metaheuristic model for optimal allocation of emergency responders given the incident occurrence rate prediction based on a statistical model and an incident severity prediction using machine learning. Specifically, the authors rely on adaptive random search (ARS) [23] for maximizing the coverage. Furthermore, within the proposed approach, a statistical model is used to predict the rate of incident occurrence, which is required for the optimal allocation of the responders. However, alongside the rate of occurrence, the severity of the incident is predicted using a machine learning model, multinomial logistic regression (MLR) [24]. Both predictions are used within the ARS-based optimization.

A further publication [25], where the authors apply metaheuristics for tuning the performance of a machine learning algorithm, proposes an approach that is aimed at decision-making in the area of building maintenance. Specifically, the authors propose the use of a machine learning technique to estimate the algal growth on a mortar surface. For this purpose, they rely on a modification of least squares support vector machines (LS-SVM) [26] that is usable for regression purposes, this modification is often referred to as least squares support vector regression (LS-SVR). Furthermore, the authors apply a metaheuristic algorithm named flower pollination algorithm (FPA) [27] for selecting optimal parameters for LS-SVR in order to achieve the optimal performance of algae growth prediction.

Another publication [28] that focuses on using metaheuristics for parameter selection of a machine learning model proposes a decision support system within a military-related area. Specifically, the authors rely on the k-nearest neighbors (KNN) machine learning algorithm for performing a classification task. In addition, a genetic algorithm is applied to select the optimal parameters of the KNN algorithm for maximizing its performance.

The following two publications can be classified as belonging to this section but also to the following [subsection 3.2](#), where approaches based on neural networks are discussed. In both cases, authors evaluate multiple types of machine learning algorithms, including traditional and those based on neural networks.

Within another recent publication [29], done in the area of manufacturing, the authors propose a method combining metaheuristics and machine learning algorithms to determine the most suitable mixture for the production of ultra-high-performance concrete. Specifically, the authors evaluate two machine learning-based regression models. Specifically, they rely on the variant of artificial neural network (ANN), called multilayer perceptron (MLP) [30], and Gaussian process regression (GPR) [31]. The models are trained on the specific experimental dataset such that these could be used to predict metrics that are critical for the evaluation of the mixture suitability given specific mix parameters. One of the models can then be selected and used as an evaluation function PSO [17], which determines the optimal parameter values for achieving a specific mix.

The authors of another publication in the manufacturing domain [32] propose a solution that combines machine learning and metaheuristic algorithms for supporting a decision on when it is worth rescheduling the production and when the delays should be tolerated. The authors use a combination of genetic algorithm (GA) and Tabu search (TS) to generate an optimal schedule. Machine learning is used to suggest a decision to reschedule or not within the specific state of the production and plan. For this purpose, authors rely on the Random Forest Classifier (RFC) [33], Support Vector Machine (SVM), Multilayer Perceptrons (MLP) [30] type of an ANN.

3.2. Neural networks and metaheuristics

In a recent publication [34], authors apply a modified version of black widow optimization (BWO) [35] for selecting features for a machine-learning-based classification with "extreme machine learning" algorithm (ELM [36]). The machine learning algorithm itself has some of its hyperparameters tuned using symbiotic organisms search algorithm (SOS) [37]. The use case of the research lies in the healthcare area and is meant to provide decision support in making the right diagnosis.

The authors of a recent publication [38] present an approach to classify traffic in a computer network with the purpose of decision management for network routing control. Specifically, the authors apply particle swarm optimization (PSO) to control the training process of an artificial neural network (ANN) for network traffic classification purposes. The authors compare the classification accuracy against two traditional machine learning baseline algorithms.

A further example of applying metaheuristics for maximizing the performance of machine learning is proposed in the manufacturing context [39]. The authors rely on the combination of a metaheuristic and a machine learning algorithm to support decision-makers in the estimation of an optimal due date in a production scheduling context. Specifically, the authors propose to use a genetic algorithm (GA) metaheuristic to decide on the architecture of an ANN-based algorithm. Furthermore, the authors suggest using an ensemble method where multiple generated architectures are used to solve the problem. Additionally, it is proposed to rely on a mechanism where multiple ANNs are employed sequentially so that each following network attempts to improve on the results of the previous one.

Similarly, metaheuristics are applied for tuning parameters of a machine learning model based on deep learning [40]. The authors rely on a long short-term memory (LSTM) deep learning model for forecasting time series representing load in an electrical network. The authors point the efficiency of the model but complexity in selecting the right hyperparameters. To tackle this challenge, they rely on GA and PSO [17] to tune certain critical hyperparameters of the selected deep learning model.

A further example of using metaheuristics as a parameterization tool for machine learning is taken in a publication [41] where the authors propose an approach for assisting decision-makers from the field of healthcare in cardiovascular disease classification. The authors rely on a machine learning model for classification based on a convolutional neural network (CNN) [42] with incorporated concepts of a Hopfield neural network (HNN) [43]. Additionally, a mayfly optimization algorithm (MF) [44] is used to select features that are used for training and inference of the machine learning model.

In [45], the authors apply a combination of machine learning and metaheuristics for the optimization of a chemical process in the production of hydrocarbon. Specifically, the authors rely on two metaheuristics: PSO [17] and grey wolf optimization (GWO) [21]. Furthermore, authors rely on a machine learning method based on ANN, specifically Multilayer Perceptron (MLP) [30]. Additionally, the authors employ a simulation of the system for data generation, which is then used to train the ANN. The ANN is trained to yield the values required for further use in the optimization process, which is done via the aforementioned PSO and GWO metaheuristics, where the ANN can be used to replace the simulator.

In the field of agriculture, the authors of a recent publication [46] propose the use of an approach based on deep neural networks (DNN) to predict the early appearance of a specific plant disease based on leaf photos. Specifically, the authors utilize a type of a DNN called Deep Residual Learning Network [47], often referred to as simply ResNet. Certain parameters of the proposed DNN are optimized using the Whale Optimization Algorithm [48] metaheuristic. The authors propose this approach for automation relying on IoT in smart farming as well as a way of gathering data for supporting decision-makers in relevant tasks.

3.3. Reinforcement learning and metaheuristics

In this section, we present the relevant works that rely on combining reinforcement learning (RL) or deep reinforcement learning (DRL) with metaheuristics. Because DRL is a subset of RL, in the majority of cases, the concepts presented for the RL algorithm are directly applicable to DRL.

In a recent publication [49], it is proposed to use an RL algorithm to orchestrate the application of multiple evolutionary metaheuristics. The authors rely on the Q-Learning [50] RL algorithm that has two actions. Each action selects one out of the two available evolutionary metaheuristics: multi-operator differential evolution (MODE) and discrete cuckoo search (DCS) [51]. When an action is selected, an iteration of the approach is performed, and the corresponding algorithm generates a set of solutions. In the next iteration, a new action is selected based on the quality and diversity of the generated solutions. Selected best solutions at each iteration can be shared between the two algorithms, MODE, and DCS, to guide the evolutionary process. The approach proposed by the authors is not industry-specific but rather attempts to tackle the general challenge of solving scheduling problems in a resource-constrained environment by improving the makespan. Therefore, we believe that this approach can potentially be applied to the smart manufacturing context.

In the field of power grid management, a publication [52] proposed to utilize DRL together with a metaheuristic algorithm as a part of a two-stage solution for grid coordination. Specifically, the authors propose to use the Weighted Delayed Deep Deterministic Policy Gradient (WD3) [53], which is based on the TD3 [54] variation of the DDPG [55] algorithm. The DRL algorithm takes demand, current various supplier pricing data, and sell prices as input. The actions produced by WD3 agent are two continuous numeric values representing power output input for various grid components. The input values of the selected DRL algorithm are generated by the genetic algorithm (GA) metaheuristic, which in turn relies on the DRL output for the fitness function. Such an approach is designed to assist the decision-making process of power grid management in terms of cost optimization and servicing demand.

Cost optimization in cloud environments is a topic of special interest for Smart Manufacturing as cloud solutions are an integral part of the Industry 4.0 concept. Authors of a recent publication [56] propose the use of a DRL algorithm together with an evolutionary metaheuristic for scheduling jobs in a cloud environment. Specifically, the authors use a genetic algorithm metaheuristic to calculate the execution order of the incoming tasks at the sub-task level. After that, a DRL approach called Deep Q Learning (DQN) [57] is used to assign each task to an existing virtual machine (VM) in a cloud environment. For each VM, the agent has an action that allows an incoming task to be assigned to the corresponding VM. The DRL agent picks the VM by assessing the incoming task and the configurations and states of the existing VMs.

3.4. Summary

In total, three types of algorithm combinations for decision support were discovered during the literature analysis. First, it's the combination of traditional (Trad.) machine learning techniques and metaheuristics. Second is the combination of metaheuristics with machine learning algorithms based on artificial neural networks (ANN) or deep neural networks (DNN). Finally, metaheuristics that are used in combination with reinforcement learning (RL) or deep reinforcement learning (DRL) algorithms are presented. The overview of the relevant publications is provided in Table 2. The table contains the summary of the specific metaheuristic algorithms (MH Alg.) and machine learning algorithms (ML Alg.) that are used in the references publications, as well as the domain of application and the nature of the used dataset.

Table 2. Literature overview.

Reference	ML Type	MH Alg.	ML Alg.	Dataset	Domain
[11]	Trad.	FA	LSSVM	Real	Disaster mgt.
[13]	Trad.	NSGA-II	XGBoost	Synthetic	Healthcare
[16]	Trad.	PSO	Regression analysis	Real	Logistics
[18]	Trad.	I-HBIA	KNN	Real	Networking
[20]	Trad.	GWO, GA	LR, DT, RF, GB, SVM, AdaBoost	Synthetic	Human resources
[25]	Trad.	FPA	LSSVR	Real	Maintenance
[22]	Trad.	ARS	MLR	Real	Incident mgt.
[28]	Trad.	GA	KNN	Real	Military
[32]	Trad. ANN	GA, TS	RFC, SVM, ANN	Real	Manufacturing
[29]	Trad. ANN	PSO	ANN, GPR	Synthetic	Manufacturing
[39]	ANN	GA	ANN	Synthetic	Manufacturing
[38]	ANN	PSO	ANN	Real	Networking
[34]	ANN	BWO, SOS	ELM	Synthetic	Healthcare
[45]	ANN	PSO, GWO	ANN	Synthetic	Manufacturing
[40]	DNN	GA, PSO	LSTM	Real	Power grid mgt.
[41]	DNN	MF	HNN, CNN	Real	Healthcare
[46]	DNN	WOA	ResNet	Real	Agriculture
[49]	RL	MODE, DCS	Q-Learning	Synthetic	—
[52]	DRL	GA	WD3	Synthetic	Power grid mgt.
[56]	DRL	GA	DQN	Synthetic	Cloud resources

4. Discussion

Overall, the majority of the discovered publications rely on various nature-inspired metaheuristic algorithms. Genetic algorithms (GA), particle swarm optimization (PSO), and their derivatives are the most often encountered ones. The selection of machine learning algorithms is more diverse. It is worth noting that artificial neural networks of various types are often applied in all manufacturing-centric cases. Out of 20 selected publications, only four are directly relevant to the area of manufacturing. Additionally, one publication is tackling challenges in the area of logistics.

An important for this literature review observation is such that, as discussed in [section 3](#), ten [34, 40, 11, 13, 39, 20, 41, 25, 28, 46] out of 20 discovered publications are, in fact, focused on optimizing the performance of machine learning and metaheuristics have a supporting role. Authors of these works rely on metaheuristics as a form of hyperparameter optimization (HPO). This means that the applied metaheuristic is not an integral part required for the decision-making process but simply a supporting mechanism for the applied ML algorithms. In most such cases, the authors apply metaheuristics to find the optimal set of parameters for maximizing the performance of the machine learning algorithm. This includes special cases such as feature selection [41] and search for an optimal artificial neural network architecture [39]. Only a single publication [39] within the area of manufacturing falls under this HPO umbrella.

The further two [32, 29] manufacturing-centric publications combining metaheuristics and machine learning use both types of algorithms to complement each other. These combinations are used to tackle the problem of production scheduling [32] or make a decision on the problem specific to the manufactured product [29]. The authors of the single publication that is representative from the area of logistics [16] also use both algorithms to complement each other.

Another notable observation that is, however, somewhat outside of the scope of interest for this literature review concerns the application of simulation in the research combination of ML and machine learning algorithms.

A discovered publication that relies on machine learning with metaheuristics and simulation comes from the area of manufacturing [45]. The simulation is primarily meant to train a machine-learning model, which in turn can be used with a metaheuristic. The idea is to replace a computationally costly simulation with a machine-learning model. Therefore, it is not a solution that actually requires all three components at all times unless the ML model requires frequent retraining. Therefore, the authors apply machine learning to estimate the parameters initially calculated using a simulation. Such estimation is faster than the direct application of the simulation, which results in a faster metaheuristic optimization calculation.

5. Future work

Given the results of the literature survey discussed in the previous sections, we propose two directions for future investigation. Firstly, the results of our literature survey indicate that a considerable amount of publications use machine learning for decision-making purposes but use metaheuristics to tune these algorithms. It is shown in a recent literature review [58], that heuristic and metaheuristic approaches are often used in the Smart Manufacturing context for this purpose. We believe that exploring the application of different classes of algorithms for this purpose may be prudent. Such as deep reinforcement learning [59], specifically in the area of decision-making in Smart Manufacturing.

Secondly, within our literature survey, no publications were discovered that explicitly apply metaheuristics together with reinforcement learning for decision-making in the manufacturing or logistics context. Therefore, we believe it is worth investigating the applicability of reinforcement learning for this case and comparing results with the existing publications.

6. Conclusion

Within this work, we conducted a systematic literature review of the solutions that assist decision-makers in various fields through a combined application of machine learning and metaheuristic algorithms. Out of 20 discovered relevant publications, only four tackle challenges in the field of manufacturing, and one on the needs of logistics. Furthermore, the majority of the discovered publications rely on two supervised and unsupervised machine learning algorithms, both traditional and those based on neural networks. Reinforcement learning is not well represented, with only three discovered publications, and none of these are specific to manufacturing.

It is important to note that ten out of 20 discovered publications merely apply metaheuristics to maximize the performance of the chosen machine learning algorithms through the algorithm parameter selection or feature engineering in the selected datasets. Formally, these fall into the scope of our literature review, but the actual decision-making support part is performed solely by the ML-based component of the algorithm combination, which relegates only a supporting role to the metaheuristics component.

The results of our literature review indicate researchers' interest in the area of combining machine learning and metaheuristics in decision-making tasks. Such combined applications take the form of either sequential use of one algorithm after another or combining the strength of both types together. However, the application of such an approach for decision-making in manufacturing and logistics is not widely represented within the results of our literature survey, despite the wide application of both ML and metaheuristics in manufacturing. Furthermore, none of the reinforcement-learning-based applications discovered in the survey are in the domain of manufacturing, thus, clearly indicating a potential for further research.

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